Faid Faisal

516-709-4261 | faidfaisal1@gmail.com | linkedin.com/in/faid-faisal | github.com/faidfaisal

Education

Stony Brook University — **Expected Graduation: May 2027**
Bachelor of Engineering in Computer Engineering – GPA: 3.88/4.0 — *Stony Brook, NY*

Relevant Coursework: Computer Architecture, Embedded Microprocessor System Design, Digital System Design, Advanced Data Structures and Algorithms, Real Time Operating Systems, Machine Learning Algorithms

Technical Skills

Languages: VHDL, SystemVerilog, Embedded C, C, C++, Python, CUDA, Assembly
ML Frameworks: PyTorch, TensorFlow, hls4ml, Scikit-learn
Tools: Vivado, Vitis, Cadence Virtuoso, Synopsys Design Compiler, QuestaSim, Altium Designer, MATLAB

Experience

Brookhaven National Laboratory — **June 2026 – August 2026**
Machine Learning Intern — *Upton, NY*

- Built and deployed machine learning classification models to autonomously categorize high-throughput X-ray scattering data from the NSLS-II CMS beamline, reducing manual data labeling overhead and accelerating experimental analysis.
- Developed an automated Python pipeline integrating Tiled, Prefect, and SciAnalysis to ingest, validate, organize, and process X-ray scattering data, enabling scalable end-to-end analysis workflows for beamline experiments.

Stony Brook University — **September 2025 – May 2026**
Undergraduate Researcher — *Stony Brook, NY*

- Developed end-to-end machine learning pipelines for automated code analysis and bug detection using AST-based program representations, libclang feature extraction, and PyTorch-based classifiers.
- Designed and evaluated graph-based deep learning architectures, including GNNs and CNNs on Code Property Graphs (CPGs), achieving superior programming error classification performance over baseline approaches.

JP Morgan Chase & CO — **February 2024 – August 2024**
Software Engineer Intern — *Manhattan, NY*

- Built Python trading dashboards with Pandas and Plotly, integrating live market data APIs to deliver real-time equity insights and improve trade decision-making by 23%.
- Developed a real-time anomaly detection system for stock price movements, backtested on 3+ years of market data and deployed into active trading workflows, improving execution accuracy by 18%.

Projects

FPGA-Based Deep Learning Accelerator for Real-Time Controls — **July 2026 – August 2026**

- Developed FPGA-based neural-network accelerators on AMD Xilinx ZCU102 platforms, integrating custom IP cores, DDR memory interfaces, DMA data paths, and low-latency hardware inference engines for real-time control and beam diagnostics applications.
- Generated and optimized ML hardware using MATLAB HDL workflows, hls4ml, Vivado, and Vitis, applying INT8 quantization and hardware-aware design techniques to balance latency, throughput, and FPGA resource utilization while validating end-to-end hardware inference.

FFT Hardware Accelerator — **May 2026 – July 2026**

- Developed FPGA-based neural-network accelerators on AMD Xilinx ZCU102 platforms, integrating custom IP cores, DDR interfaces, DMA engines, and streaming inference pipelines for real-time control and beam diagnostics.
- Generated and optimized ML hardware using MATLAB HDL workflows, hls4ml, Vivado, and Vitis, applying INT8 quantization and hardware-aware design techniques to maximize throughput and resource efficiency while validating end-to-end deployment.

Four-Stage Pipelined SIMD Multimedia Processor — **August 2025 – December 2025**

- Designed a four-stage SIMD processor in VHDL with a 128-bit ALU, 32x128-bit register file, and hazard-free forwarding unit executing up to 64 25-bit instructions.
- Built a Python assembler and cycle-accurate testbench to validate forwarding behavior across the full IF/ID/EX/WB pipeline.